Entropy Walls and Sphaleron Windows: Baryogenesis in Modulus-Dominated Universes and the Large Volume Scenario

Vishal Hiralal Thakur
ORCID 0009-0008-8406-0508 beingcosmic44@gmail.com

Abstract

Light moduli generically dominate the energy density before they decay, and the entropy released dilutes any earlier baryon asymmetry. We ask which baryogenesis mechanisms can operate in such a universe, and apply the answer to the Large Volume Scenario (LVS). For an early matter-dominated era (EMDE) we compute the temperature window in which electroweak sphalerons are in equilibrium, using the lattice rate, and the efficiency with which baryon- or lepton-number-violating decays are converted into a final asymmetry. For axion-driven mechanisms, the kinetic energy of the axion bounds the Chern–Simons chemical potential and gives $F \lesssim 4 \times 10^{10}$ GeV for the anomalous decay constant; the bulk and fibre axions of LVS have $F \sim 10^{16}$ GeV. In explicit Swiss-cheese and K3-fibred vacua, electroweak baryogenesis before the decay requires $\mathcal{V} \lesssim 4 \times 10^7$ and 7×10^6, respectively, and leptogenesis from modulus decay $\mathcal{V} \lesssim 10^7$ and 2×10^6, so TeV-scale gravitino masses are excluded in the absence of thermal inflation. Baryon-violating decays remain possible. For sequestered soft terms, $m_{\rm soft} \sim M_{\rm Pl}/\mathcal{V}^2$, TeV-scale soft masses correspond to $\mathcal{V} \sim 10^7$–10^8, and the observed asymmetry then requires a CP-violating baryon number per decay $\epsilon \simeq (1\text{–}6) \times 10^{-4}$. The gaugino-pair channel on which the standard moduli-induced mechanism relies is closed in LVS: it is kinematically forbidden for gravity-mediated gaugino masses and vanishes at tree level when the gauge kinetic function is independent of the volume moduli. The Higgs and higgsino channels, with branching ratio up to about 0.4, would then have to supply a CP asymmetry per decay of $(1.6\text{–}7) \times 10^{-4}$; for a higgsino decaying through a gluino loop the asymmetry vanishes at leading order for a pure higgsino and reaches the required size only for a bino admixture of at least 25%. Affleck–Dine baryogenesis from a baryonic flat direction reproduces the observed asymmetry for $\mathcal{V} \lesssim 6 \times 10^5$ if the operator scale is the Planck scale, but the flat direction is then displaced beyond the string scale. Requiring the displacement to lie within the effective theory lowers the maximal volume to about 10^4.

1 Introduction

String compactifications contain moduli that are light compared with the string scale and couple to matter only through Planck-suppressed interactions. Their coherent oscillations come to dominate the energy density before decay: the cosmological moduli problem [1, 2, 3]. In the Large Volume Scenario [4, 5] the volume modulus has mass $m \sim M_{\rm Pl}\mathcal{V}^{-3/2}$ and the gravitino mass is $m_{3/2} \sim M_{\rm Pl}/\mathcal{V}$. TeV-scale soft terms need $\mathcal{V} \sim 10^{14}$–10^{15}, for which m is of order 10–100 keV and the modulus decays after BBN unless it is diluted [6].

Three facts constrain baryogenesis in a modulus-dominated universe. We call the suppression of any pre-decay asymmetry by the entropy released in the decay the *entropy wall*, and the range of temperatures in which electroweak sphalerons are in equilibrium the *sphaleron window*. First,

an asymmetry produced before the decay is diluted by the entropy released. Second, the Hubble rate during the EMDE is far larger than in radiation domination at the same temperature, so electroweak sphalerons may or may not be in equilibrium when one expects. Third, an asymmetry sourced by a CP-odd background is limited by how fast that background can move. The aim of this paper is to make each statement quantitative and to follow the consequences for LVS.

Results. (1) The sphaleron-equilibrium window in an EMDE is computed with the lattice rate; it closes for $T_R \lesssim 2.6$ MeV (Sec. 3). (2) Baryon-violating decays are converted with efficiency $\eta_B \simeq 1$ for $T_R \lesssim 30$ GeV, while lepton-violating decays are converted with $\eta_L \propto T_R^4$ (Sec. 3.3). (3) Any axion-driven, sphaleron-mediated mechanism obeys $F \lesssim 4.2 \times 10^{10}$ GeV (Sec. 4). (4) For asymmetries generated at temperature $T_{\rm gen}$ before the decay, $T_R \gtrsim 6\,{\rm GeV}\,(T_{\rm gen}/131.7\,{\rm GeV})$ (Sec. 5). (5) These requirements are translated into bounds on $\mathcal{V}$ in explicit Swiss-cheese and K3-fibred vacua, and, for sequestered soft terms, into the CP asymmetry that baryon-violating modulus decays must supply and its comparison with the known upper range of $\epsilon_{\rm CP}$ (Sec. 6). (6) The Affleck–Dine mechanism is analysed with attention to the validity of the effective field theory (Sec. 7).

Relation to earlier work. The following are known and used here. The law $T \propto a^{-3/8}$ and the enhanced Hubble rate during reheating [13]; the existence of a maximal temperature $T \lesssim (\alpha_W^4 M_{\rm Pl} T_R^2)^{1/3}$ for sphaleron equilibrium and the dilution $Y_B \propto (T_R/T_c)^5$ [10, 12]; the suppression of lepton asymmetries produced at low reheating temperature [11]; the relation between axion rotation, PQ charge and baryon number [14, 15, 16]; and the cosmology of LVS [6, 7, 8]. Affleck–Dine baryogenesis followed by dilution from moduli decay was studied in [19], and in LVS in [20]; baryogenesis from moduli decay in [21] and, through gluino decays with $Y_B = Y_{\tilde g}\,\epsilon_{\rm CP}$, in [22]; baryogenesis in an EMDE with sphaleron washout in [23]. What we add is: the lattice-rate coefficients of the sphaleron window and the closed forms in Sec. 3.3; the multi-axion form of the kinetic bound, including its dark-matter-constrained version; the channel-by-channel volume bounds in explicit vacua; and a check of the Affleck–Dine analysis against the validity of the effective theory, which changes the conclusion of the naive estimate.

Throughout, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV is the reduced Planck mass and $Y_B^{\rm obs} = 8.6 \times 10^{-11}$.

2 Moduli domination and entropy production

Let the lightest long-lived modulus ϕ, of mass m, decay with width Γ to radiation:

$$\dot\rho_\phi + 3H\rho_\phi = -\Gamma\rho_\phi, \qquad \dot\rho_R + 4H\rho_R = \Gamma\rho_\phi, \qquad H^2 = \frac{\rho_\phi + \rho_R}{3M_{\rm Pl}^2}. \tag{1}$$

We define T_R by $\Gamma = H_{\rm RD}(T_R)$,

$$\Gamma = \sqrt{\frac{\pi^2 g_R}{90}}\,\frac{T_R^2}{M_{\rm Pl}}, \qquad g_R \equiv g_*(T_R). \tag{2}$$

We assume that ϕ dominates before it decays, as is generic for an initial displacement of order $M_{\rm Pl}$; for a small displacement the dilution is absent but BBN and dark radiation constrain the model, and our bounds are for the dominating case.

For $H \gg \Gamma$ the radiation reaches a quasi-stationary state with $\rho_R \propto a^{-3/2}$. Then $(4-\frac{3}{2})H\rho_R = \Gamma\rho_\phi$, so $\rho_R = \frac{6}{5}\Gamma M_{\rm Pl}^2 H$ and

$$H(T) = \frac{\pi^2 g_*(T)}{36}\frac{T^4}{\Gamma M_{\rm Pl}^2} = \kappa\,\frac{T^4}{T_R^2 M_{\rm Pl}}, \qquad \kappa = \frac{\pi^2 g_*(T)/36}{\sqrt{\pi^2 g_R/90}}, \qquad T \propto a^{-3/8}. \tag{3}$$

For $g_* = 106.75$, $g_R = 10.75$ the coefficient exceeds the usual $\sqrt{\pi^2 g_*/90}$ by a factor 7.9. We have checked Eq. (3) against numerical solutions of Eq. (1) at $T = 131.7$ GeV to better than 0.1%.

Dilution. The comoving entropy $S = sa^3$ scales as T^{-5} in this regime. An asymmetry frozen into the comoving volume at radiation temperature $T_{\rm gen}$ is reduced by

$$\mathcal{D} \equiv \frac{S(T_{\rm gen})}{S_{\rm fin}} = \mathcal{C}\left(\frac{T_R}{T_{\rm gen}}\right)^5, \qquad \mathcal{C} \simeq 0.16\,\frac{g_R}{g_*(T_{\rm gen})}. \tag{4}$$

The coefficient follows from matching the sudden-decay entropy to Eq. (3) (Appendix A); it gives 0.016 for $T_R \lesssim 50$ MeV and 0.11–0.14 for $0.2 \lesssim T_R \lesssim 60$ GeV, in agreement with the numerical solution (Fig. 1). For $T_R = 18$ MeV, $\mathcal{D} = 7.6 \times 10^{-22}$.

Validity: primordial radiation. Equation (3) assumes that the radiation at $T_{\rm gen}$ is produced by the decays. If the modulus starts to oscillate at $H = m$ in a bath of primordial radiation left by the inflaton, that radiation redshifts as a^{-1} in temperature and dominates the temperature until it falls below the decay product, which cools as $a^{-3/8}$. With $\rho_\phi = \rho/6$ at $H = m$, the crossover occurs at

$$T_\times \simeq 4.5 \times 10^2\ {\rm GeV}\left(\frac{m}{10^7\ {\rm GeV}}\right)^{13/10}. \tag{5}$$

The steady-state description at $T_{\rm gen} = 131.7$ GeV requires $T_\times \gtrsim T_{\rm gen}$, that is $m \gtrsim 4 \times 10^6$ GeV or $T_R \gtrsim 1$ GeV. All bounds quoted below that rely on $T_R \gtrsim 6$ GeV satisfy this. For smaller T_R the numbers in Table 1 and Fig. 2 depend on the pre-oscillation history; the qualitative conclusions do not, since in that case the true $H(T_*)$ is smaller and $\Gamma/H(T_*) \ll 1$ remains.

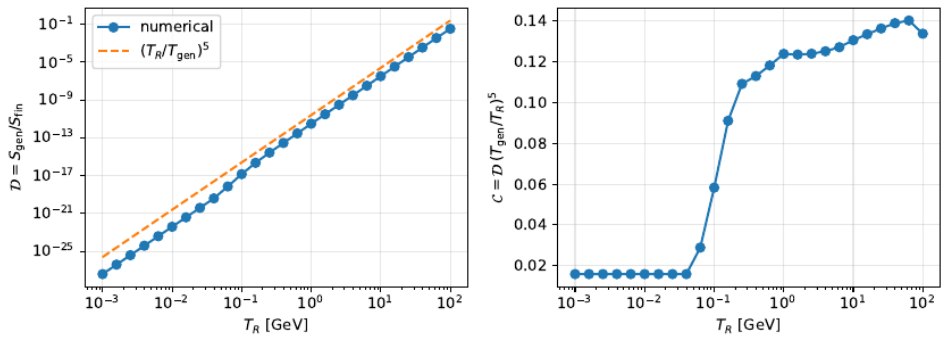


Figure 1: Entropy dilution $\mathcal{D} = S(T_{\rm gen})/S_{\rm fin}$ for $T_{\rm gen} = 131.7$ GeV from Eq. (1) (left) and the prefactor $\mathcal{C}$ (right). The step near $T_R \sim 0.1$ GeV is the change of g_* across the QCD crossover in our approximate $g_*(T)$.

3 Electroweak sphalerons in an EMDE

3.1 Susceptibility

A sphaleron transition changes Chern–Simons number by $\Delta N_{\rm CS} = \pm 1$ and produces twelve left-handed doublet fermions. A bias $\mu_{\rm CS}$ conjugate to $N_{\rm CS}$ enforces $n_g(3\mu_Q + \mu_L) = \mu_{\rm CS}$. Imposing Yukawa equilibrium ($\mu_u = \mu_Q + \mu_H$, $\mu_d = \mu_Q - \mu_H$, $\mu_e = \mu_L - \mu_H$), hypercharge neutrality and $B - L = 0$, one finds (Appendix B)

$$n_B = \tfrac{12}{79} T^2 \mu_{\rm CS} \quad \Rightarrow \quad Y_B = \xi\, \frac{\mu_{\rm CS}}{T}, \qquad \xi = \frac{12/79}{2\pi^2 g_*/45} = 3.24 \times 10^{-3}. \tag{6}$$

With $\mu_{\rm CS} = 0$ the same system gives the standard $B = \frac{28}{79}(B - L)$. In an EMDE the Hubble rate at $T \sim 150$ GeV is enhanced by $7.9\, T^2/T_R^2$ relative to radiation domination, and the electron Yukawa coupling is out of equilibrium for $T_R \lesssim 15$ GeV. Removing e_R from the equilibrium network lowers ξ by 4%.

3.2 Rate and equilibrium window

The lattice rate for $m_H = 125$ GeV is [18]

$$\frac{\Gamma_{\rm sph}}{T^4} = \begin{cases} \exp(0.83\, T/{\rm GeV} - 147.7), & 130\ {\rm GeV} < T < T_c, \\ 18\, \alpha_W^5, & T > T_c, \end{cases} \qquad T_c = 159\ {\rm GeV}. \tag{7}$$

In radiation domination sphalerons freeze out at $T_* = 131.7 \pm 2.3$ GeV, where the washout rate $\lambda_{\rm sph} = c^{-1}\Gamma_{\rm sph}/T^3$ equals H. We fix $c = 0.12$ (i.e. $1/c \simeq 8.5$) to reproduce this value; it is a calibration, not a prediction, and we vary it by a factor of ten below. In an EMDE H is given by Eq. (3) and grows as T^4, whereas $\Gamma_{\rm sph}/T^3$ grows only as T above T_c. Equilibrium therefore holds only in a window

$$T_*(T_R) < T < T_{\rm up}(T_R), \qquad T_{\rm up}^3 = \frac{18\, \alpha_W^5\, T_R^2 M_{\rm Pl}}{c\, \kappa}. \tag{8}$$

The scaling $T_{\rm up} \propto T_R^{2/3}$ is due to [10]; the lattice rate fixes the coefficient and the broken-phase fit fixes the lower edge (Table 1). The window closes when $T_{\rm up} = T_c$, i.e.

$$T_R^{\rm eq} = \left(\frac{c\, \kappa\, T_c^3}{18\, \alpha_W^5 M_{\rm Pl}} \right)^{1/2} = 2.6\ {\rm MeV} \tag{9}$$

(0.8–8.1 MeV for $c \to c/10,\ 10c$), at or below the BBN bound. The fit (7) is discontinuous at T_c by a factor ~ 5, so edges near T_c are uncertain by a few GeV.

T_R	18 MeV	56 MeV	1 GeV	10 GeV
T_* [GeV]	156	153	145	139
$T_{\rm up}$ [GeV]	582	1296	$\gtrsim 3000$	$\gtrsim 3000$

Table 1: Sphaleron equilibrium window in an EMDE.

3.3 Washout and conversion efficiencies

Let each modulus decay create baryon number ϵ_B and lepton number ϵ_L. With comoving densities $N_X = n_X a^3$, $N_\phi = \rho_\phi a^3/m$,

$$\dot{N}_{B-L} = (\epsilon_B - \epsilon_L)\Gamma N_\phi, \qquad \dot{N}_B = \epsilon_B \Gamma N_\phi - \lambda_{\rm sph}(T)\Big(N_B - \tfrac{28}{79} N_{B-L}\Big), \tag{10}$$

to be solved with Eq. (1). We use the symmetric-phase target $\frac{28}{79}$; the broken-phase correction near freeze-out is a few percent. The system is linear in the sources, so $N_B^{\rm fin} = \eta_B \epsilon_B N_\phi^{\rm tot}$ for $\epsilon_L = 0$ and $N_B^{\rm fin} = -\eta_L \epsilon_L N_\phi^{\rm tot}$ for $\epsilon_B = 0$, with $\eta_{B,L}$ depending only on T_R (Fig. 2).

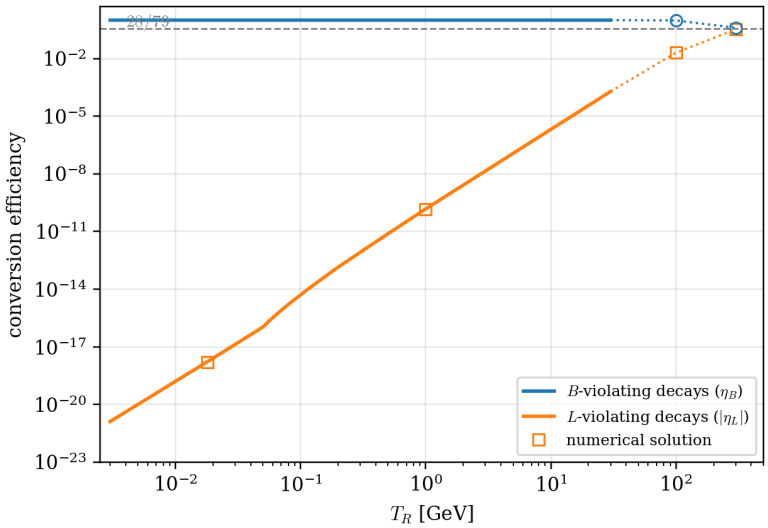


Figure 2: Conversion efficiencies. Curves: closed forms (11) for $T_R \le 30$ GeV; open markers: numerical solutions of Eq. (10); dotted lines join the numerical values at larger T_R. Baryon-violating decays survive for $T_R \lesssim 30$ GeV; lepton asymmetries are converted only if an appreciable fraction of the decays occurs while sphalerons are in equilibrium.

Closed forms. For $T_R \ll T_*$ the fraction of the modulus energy that has decayed by $t = 2/3H$ is $\Gamma t = \frac{2}{3}\Gamma/H$. Lepton number produced before sphaleron freeze-out is converted with coefficient 28/79, and all later decays leave B untouched. Using Eqs. (2) and (3), $\Gamma/H = \frac{2g_R}{5g_*}(T_R/T)^4$, so

$$\eta_L \simeq \frac{28}{79}\cdot\frac{2}{3}\frac{\Gamma}{H(T_*)} = \frac{28}{79}\,\frac{4}{15}\,\frac{g_R}{g_*}\left(\frac{T_R}{T_*}\right)^4, \qquad \eta_B \simeq 1 - \frac{51}{79}\cdot\frac{2}{3}\frac{\Gamma}{H(T_*)}. \tag{11}$$

Here $T_* = T_*(T_R)$ is the lower edge of the window from Table 1; with it, Eq. (11) gives 1.7×10^{-18}, 1.5×10^{-10}, 2.0×10^{-6} at $T_R = 18$ MeV, 1 GeV, 10 GeV, within 10% of the numerical solution. (Using the radiation-domination value $T_* = 131.7$ GeV overestimates η_L by up to a factor 2.) Numerically $\eta_B = 1$ to better than 10^{-3} for $T_R \lesssim 30$ GeV, falling to 0.96 at 100 GeV and 0.40 at 300 GeV; $\eta_L \simeq 1.4\times10^{-10}(T_R/\text{GeV})^{4.1}$ for 1–10 GeV, reaching 0.02 at 100 GeV and 0.33 at 300 GeV. Leptogenesis from modulus decay therefore needs $T_R \gtrsim$ 100–300 GeV; for $T_R \lesssim 30$ GeV the sphalerons are spectators. Multiplying Eq. (11) by the decay yield $\epsilon\kappa_Y T_R/m$ gives $Y_B \propto \epsilon T_R^5/(T_*^4 m)$, the scaling found for non-thermal leptogenesis in [11].

4 Axion-driven sources

Let $\theta^i = a^i/f$ have kinetic term $\frac{1}{2}G_{ij}\dot\theta^i\dot\theta^j$ and couple through $\mathcal{L} \supset \frac{g^2}{32\pi^2}c_i\theta^i W\tilde W$ with constant c_i. In string compactifications $c_i = k_i$ are the integers in $f_W = k_i T^i$. Since $\frac{g^2}{32\pi^2}W\tilde W = \partial_\mu K^\mu$,

$$\mu_{\rm CS} = \frac{d}{dt}(c_i\theta^i) = c_i\dot\theta^i. \tag{12}$$

Remark (frame independence). $c_i\theta^i$ is a scalar, so $\mu_{\rm CS}$ is unchanged by any time-dependent rotation $R(t)$ of the field-space basis: $\tilde c_\alpha\dot{\tilde\theta}^\alpha + \dot{\tilde c}_\alpha\tilde\theta^\alpha = c_i\dot\theta^i$. In particular a rotating eigenbasis, such as an avoided crossing in a thermal mass matrix, produces no chemical potential unless the axions acquire a coordinate velocity.

Define $F \equiv (c_i G^{ij} c_j)^{-1/2}$; for one axion with $\theta = a/f$, $F = f/N$.

Bound. If sphalerons are in equilibrium at $T_{\rm gen}$ and $\rho_{\rm kin} = \frac{1}{2}G_{ij}\dot\theta^i\dot\theta^j \le \rho_{\rm tot} = 3H^2M_{\rm Pl}^2$, the Cauchy–Schwarz inequality in the metric G gives $|c_i\dot\theta^i| \le F^{-1}\sqrt{2\rho_{\rm kin}}$, hence

$$Y_B^{\rm gen} \le \xi\,\frac{\sqrt{6}\,H(T_{\rm gen})M_{\rm Pl}}{F\,T_{\rm gen}} \xrightarrow{\rm RD} \xi\,\pi\sqrt{\frac{g_*}{15}}\frac{T_{\rm gen}}{F} \simeq \frac{3.6\ {\rm GeV}}{F}, \qquad F \le F_{\rm max} = 4.2\times10^{10}\ {\rm GeV}. \tag{13}$$

Only the sphaleron freeze-out region contributes, because $B+L$ relaxes to its equilibrium value while sphalerons are active and $B-L$ is not generated by $\mu_{\rm CS}$; thus $T_{\rm gen} = T_*(T_R)$. The bound uses no information about the source and is saturated only if the axion kinetic energy equals the total energy density (kination). In an EMDE, $Y_B^{\rm fin} \propto H_{\rm EMDE}\mathcal{D} \propto T_R^3$, so an EMDE only weakens the mechanism: for $T_R \le 30$ GeV the maximal yield is lower by 4.9×10^{-3}, $F_{\rm max} \to 2\times10^8$ GeV. A source limited by the Hubble rate, $|c_i\dot\theta^i| \le \nu H$, gives $Y_B \le \xi\nu H/T$, which at $T = 131.7$ GeV requires $\nu \gtrsim 10^8$.

For a single axion, with $n_\theta = f^2\dot\theta$ and $Y_\theta = n_\theta/s$, the same susceptibility gives $Y_B = \frac{12}{79}N(T^2/f^2)Y_\theta$, the relation underlying axiogenesis [14].

Dark-matter-constrained version. If the rotation ends as cold axion dark matter with conserved PQ charge, $m_a Y_\theta^{\rm fin} \le \rho_{\rm DM}/s = 4.4\times10^{-10}$ GeV. Baryons and axions are diluted equally, so for any EMDE history

$$\frac{f^2 m_a}{N} \le \frac{(12/79)\,T_{\rm gen}^2(\rho_{\rm DM}/s)}{Y_B^{\rm obs}} = 1.35\times10^4\ {\rm GeV}^3. \tag{14}$$

For the QCD axion, $m_a f \simeq 5.7\times10^{-3}$ GeV^2 gives $f \lesssim 2\times10^6 N$ GeV. This is more restrictive than the values 10^7–10^8 GeV quoted in the axiogenesis literature. The bound can be evaded by dissipating the rotation [16], which is why Eq. (13), which assumes nothing about the endpoint, is the more robust statement.

Classification. Let $H_* = H(T_*)$. (A) $m_a \lesssim H_*$: the field is frozen, $|\dot\theta| \lesssim H_*\theta$, and the Hubble-limited bound applies. (B) $m_a \gg H_*$, $\Gamma_a \ll H_*$: the axion oscillates and is long lived; Eq. (13) applies. (C) $\Gamma_a \gtrsim H_*$: the oscillation is suppressed by $e^{-\Gamma_a/4H_*}$ and no coherent velocity survives at freeze-out.

5 Generic bound on T_R for pre-decay generation

Let a mechanism act at $T_{\rm gen}$ with maximal chemical potential $(\mu/T)_{\max}$. Then $Y_B^{\rm fin} \leq \mathcal{D}(T_R)\,\xi\,(\mu/T)_{\max}$, and

$$T_R \geq T_R^{\min} \simeq T_{\rm gen}\left[\frac{Y_B^{\rm obs}}{\mathcal{C}\,\xi\,(\mu/T)_{\max}}\right]^{1/5}. \tag{15}$$

For $T_{\rm gen} = 131.7$ GeV: $T_R^{\min} = 6.1$, 9.6, 15.1, 23.9, 37.7 GeV for $(\mu/T)_{\max} = 1, 10^{-1}, \ldots, 10^{-4}$. Since $T_R^{\min} \propto T_{\rm gen}$, mechanisms operating at lower temperature are less constrained; the bound applies to generation at the electroweak scale or above. In terms of the modulus mass, with $\Gamma = c_\Gamma m^3/4\pi M_{\rm Pl}^2$ and $c_\Gamma = 1$, $T_R \geq 6.1$ GeV is $m_\Phi \gtrsim 1.5 \times 10^7$ GeV (scaling as $c_\Gamma^{-1/3}$), about two orders of magnitude above the value $m_\Phi \gtrsim 1.2 \times 10^5$ GeV required by BBN.

6 Application to the Large Volume Scenario

6.1 Explicit vacua

We use the Swiss-cheese compactification on $\mathbb{P}^4_{[1,1,1,6,9]}$ ($h^{1,1} = 2$, $\chi = -540$) [4], with

$$K = -2\ln\!\big(\mathcal{V} + \tfrac{\hat{\xi}}{2}\big), \quad \mathcal{V} = \gamma(\tau_b^{3/2} - \tau_s^{3/2}), \quad W = W_0 + Ae^{-aT_s}, \tag{16}$$

$\gamma = 1/9\sqrt{2}$, $\hat{\xi} = -\zeta(3)\chi/2(2\pi)^3 g_s^{3/2}$, $W_0 = |A| = 1$, $g_s = 0.3$ (0.4 for the smallest volumes), $a = 2\pi/N$ treated as continuous. The exact $\mathcal{N} = 1$ potential is minimised at high precision and the Hessian checked to be positive definite. The vacua are AdS, which changes masses at $O(1)$ relative to a de Sitter vacuum. Ten vacua cover $\mathcal{V} = 2.5 \times 10^3$–$10^{16}$. They confirm $m_{3/2} \simeq M_{\rm Pl}/\mathcal{V}$, $m_{\tau_b} = c_m M_{\rm Pl}\mathcal{V}^{-3/2}$ with $c_m = 1.0$–2.1, and $m_{\tau_s} \simeq m_\rho \simeq (17$–$78)\,m_{3/2}$; a modulus mass of 10^5 GeV corresponds to $\mathcal{V} \sim 10^9$, not 10^{15}. We take $\Gamma = c_\Gamma m^3/4\pi M_{\rm Pl}^2$ with $c_\Gamma = 1$ (varied between 0.3 and 8). The width to the bulk axion partner, $m^3/48\pi M_{\rm Pl}^2$ [7], produces dark radiation; $\Delta N_{\rm eff} < 0.3$ requires $c_\Gamma \gtrsim 0.9$–1.8 [8].

For the K3-fibred model [9] the fibre modulus is the lightest. With the string-loop potential of [9], which is conjectural, we find $m_f = c_f M_{\rm Pl}\mathcal{V}^{-5/3}$ with $c_f = 1.45$ for the reference loop data (0.70–3.0 under factor-of-three variations; 17 for the published TeV-scale parameter set), so $T_R \propto M_{\rm Pl}\mathcal{V}^{-5/2}$. For the published LV set the numerical fibre mass reproduces the analytic $m_f^2 = 4(1 + \frac{7}{6}R)C_2^2/\mathcal{V}^{10/3}$ to fourteen digits. If heavier moduli decay earlier, additional dilution can only reduce the final asymmetry, so the bounds below hold for multi-stage histories.

Maximal $\mathcal{V}$ for	$T_R \geq 6.1$ GeV	$T_R \geq 100$ GeV	$T_R \geq 4$ MeV
Swiss-cheese	3.8×10^7	1.1×10^7	1.2×10^9
K3-fibred, $c_f = 1.45$	6.6×10^6	2.1×10^6	1.5×10^8
K3-fibred, $c_f = 0.7$–3.0	4.3×10^6–1.0×10^7	1.4×10^6–3.3×10^6	9.8×10^7–2.4×10^8

Table 2: Upper bounds on $\mathcal{V}$ from pre-decay electroweak baryogenesis, decay leptogenesis, and BBN.

6.2 Channel by channel

Pre-decay electroweak baryogenesis requires $T_R \gtrsim 6$–38 GeV, i.e. $\mathcal{V} \lesssim 3.8 \times 10^7$ (Swiss-cheese) and 7×10^6 (K3), $m_{3/2} \gtrsim 10^{10}$ GeV. At $\mathcal{V} = 10^{16}$, $T_R \simeq 6 \times 10^{-19}$ GeV.

Axion-driven baryogenesis. For $f_W = NT$ with T a bulk or fibre modulus, $K_{T\bar{T}} = O(1)/\tau^2$ and $\alpha_W^{-1} = N\tau$, so

$$F = \frac{\kappa_F\, \alpha_W M_{\rm Pl}}{4\pi}, \qquad \kappa_F = 2\tau\sqrt{2K_{T\bar{T}}} = O(1)\ (\sqrt{2}\ \text{for the fibre}), \tag{17}$$

independent of N and τ: $F \simeq 9.3\times 10^{15}$ GeV, 2×10^5 times the bound (13), even at kination. For the Standard Model on a blow-up cycle, $K_{s\bar{s}} \simeq 3\gamma/8\sqrt{\tau_s}\mathcal{V}$ and $F_s \simeq 4.0\times 10^{16}\,{\rm GeV}/\sqrt{\mathcal{V}}$, which satisfies (13) for $\mathcal{V} \gtrsim 9\times 10^{11}$. But the axion is heavy, $m_\rho \simeq (17\text{–}78)m_{3/2}$, and decays to gauge bosons with $\Gamma_\rho \simeq \frac{3}{8}\alpha_W^2 m_\rho^3/32\pi^3 F_s^2$; in the explicit vacua $\Gamma_\rho/H(T_*) \geq 2.3\times 10^3$ for all $\mathcal{V}$, so it is of type (C) and provides no velocity. (For $\mathcal{V} \gtrsim 10^{10}$ the radiation at T_* is not in the steady state, but there T_R is far below BBN in any case.)

Lepton-violating decays. $\eta_L \simeq 1.4\times 10^{-10}(T_R/{\rm GeV})^{4.1}$, so $T_R \gtrsim 100$ GeV is needed: $\mathcal{V} \lesssim 1.1\times 10^7$.

Baryon-violating decays. With ϵ the CP-violating baryon number per decay, $Y_B = \epsilon\,\eta_B\kappa_Y T_R/m$ with $\kappa_Y \simeq 0.7$–0.9. Since $T_R/m \propto \mathcal{V}^{-3/4}$, $\mathcal{V}_{\rm max} \propto \epsilon^{4/3}$; in the explicit vacua $\mathcal{V}_{\rm max} = 3.1\times 10^{12}$, 7.3×10^9, 2.6×10^6, 6.5×10^3 for $\epsilon = 1$, 10^{-2}, 10^{-4}, 10^{-6}. TeV-scale LVS requires $\epsilon \gg 1$. The value of ϵ is model dependent; mechanisms in which it is fixed, through the decay chain modulus $\to$ gaugino pair $\to$ *udd*, are described in [21, 22] for KKLT-type and other stabilisation schemes; the LVS case is considered next.

6.3 Sequestered soft terms and the required CP asymmetry

In sequestered LVS the soft terms on the Standard Model brane are suppressed relative to $m_{3/2} \sim M_{\rm Pl}/\mathcal{V}$ by at least $\mathcal{V}^{-1/2}$ and may be as small as $M_{\rm Pl}/\mathcal{V}^2$; gaugino masses are of order $m_{3/2}/\mathcal{V}$, and scalar masses are either comparably small or larger by $\mathcal{V}^{1/2}$ [17]. For TeV soft terms [17] quotes $\mathcal{V} \sim 10^6$–10^7; the precise value depends on W_0 and g_s. For the vacua used here ($W_0 = 1$), $m_{\rm soft} = M_{\rm Pl}/\mathcal{V}^2$ lies between 10 TeV and 0.1 TeV for $1.5\times 10^7 \lesssim \mathcal{V} \lesssim 1.5\times 10^8$. In this range $m_{3/2} \sim 10^{10}$–10^{11} GeV, $m_{\tau_b} \simeq 2\times 10^6$–$6\times 10^7$ GeV and $T_R \simeq 0.4$–50 GeV, so that $\eta_B \simeq 1$ (Sec. 3.3), and decays to superpartners are kinematically open. Solving $Y_B = \epsilon\,\eta_B\kappa_Y T_R/m$ for ϵ gives (Fig. 4)

$$\epsilon \simeq 1.3\times 10^{-4},\ 2.8\times 10^{-4},\ 5.6\times 10^{-4} \qquad \text{at } \mathcal{V} = 1.5\times 10^7,\ 5\times 10^7,\ 1.5\times 10^8, \tag{18}$$

with a minimum $\epsilon \simeq 2\times 10^{-4}$ near $\mathcal{V} = 3\times 10^7$.

This requirement is to be compared with what the decay chain can supply. In the mechanism of [22] the modulus decays to a gaugino pair, and the gluino decays to tsb through $W \supset \lambda''_{323}U^cD^cD^c$ with a CP asymmetry from the interference of tree and loop diagrams; in our normalisation $\epsilon = \frac{4}{3}\,{\rm Br}\,\epsilon_{\rm CP}$, with $\epsilon_{\rm CP} \lesssim 3\times 10^{-4}\sin 2\phi_{12}$ for the MSSM spectra of [22]. The branching ratios follow from the supergravity data. We have computed them in the vacua of Sec. 6 using the decay widths of [22], with the light eigenstate Φ of $G^{-1}\partial^2 V$ normalised canonically and $\delta\tau^i = v^i\phi$. The width to the massless axion of T_b is

$$\Gamma_{aa} = \frac{g^2 m^3}{128\pi M_{\rm Pl}^2}, \qquad g = \frac{\partial_\phi K_{b\bar{b}}}{K_{b\bar{b}}}, \qquad g^2 = \frac{8}{3}, \tag{19}$$

that is $\Gamma_{aa} = m^3/48\pi M_{\rm Pl}^2$, which we confirm numerically to four digits in each vacuum. For the Higgs sector, with $Z \propto \mathcal{V}^{-2/3}$ one finds $\partial_\phi \ln Z = \sqrt{2/3}$ in every vacuum, so that

$$\Gamma_{HH} = \Gamma_{\tilde{h}\tilde{h}} = 2z^2\,\Gamma_{aa}, \qquad z \equiv Z/\hat{Z}, \tag{20}$$

and $\mathrm{Br}_{\tilde{h}\tilde{h}} = 2z^2/(1+4z^2+\Gamma_{gg}/\Gamma_{aa})$, which is 0.40 for $z=1$ and 0.25 for $z=1/2$. For gauge bosons and gauginos two cases arise. If the gauge kinetic function is $f = T_s$, as for a stack wrapping the blow-up cycle, then $\Gamma_{gg}/\Gamma_{aa} = 0.03$–$0.06$ (Table 3) and the gaugino mass $M_{\tilde{\lambda}} = F^{T_s}/2\tau_s$ exceeds m_Φ by a factor 17–630, which grows as $\mathcal{V}^{1/2}$; the gaugino-pair channel is closed. If f does not depend on T_b or T_s, as for D3-branes at a singularity at leading order, then $\partial_\Phi f = 0$ and Γ_{gg} and $\Gamma_{\tilde{\lambda}\tilde{\lambda}}$ vanish at tree level; the sequestered gaugino masses of order $m_{3/2}/\mathcal{V}$ lie below m_Φ, but the corresponding couplings arise only at loop level and from anomaly-mediated terms, which are further suppressed. In neither case is $\mathrm{Br}_{\tilde{\lambda}\tilde{\lambda}}$ of order one, so the gluino chain of [22] does not operate in LVS.

$\mathcal{V}$	m_Φ [GeV]	$M_{\tilde{\lambda}}/m_\Phi$ $(f=T_s)$	Γ_{gg}/Γ_{aa} $(f=T_s)$	$(\Gamma_{HH}+\Gamma_{\tilde{h}\tilde{h}})/\Gamma_{aa}$
2.6×10^5	3.0×10^{10}	17	0.063	$4z^2$
3.4×10^6	6.0×10^8	54	0.045	$4z^2$
1.2×10^7	8.3×10^7	99	0.039	$4z^2$
4.5×10^7	1.1×10^7	183	0.034	$4z^2$
1.7×10^8	1.6×10^6	339	0.029	$4z^2$
6.1×10^8	2.1×10^5	630	0.026	$4z^2$

Table 3: Light-modulus decay widths in the explicit Swiss-cheese vacua ($W_0 = A = 1$, $g_s = 0.3$, no uplift), from the formulas of [22] with $N_g = 12$.

The baryon asymmetry must then be carried by the Higgs sector. In the pattern of the hidden-gaugino variant of [22], in which a fermion decays through a gluino loop with ϵ_{CP} up to 10^{-2}, a higgsino pair with $\mathrm{Br}_{\tilde{h}\tilde{h}} \lesssim 0.4$ requires $\epsilon_{\mathrm{CP}} \simeq (1.6\text{–}7)\times10^{-4}$ across the window. We have computed this asymmetry by an explicit amplitude calculation (Appendix D). The neutral higgsino couples to the top and the stop through $W \supset y_t H_u Q U^c$, with the structures $(\tilde{h}\psi_Q)\tilde{t}_R^*$ and $(\tilde{h}\psi_{U^c})\tilde{t}_L$, each with coefficient $y_t/\sqrt{2}$ for a Majorana mass eigenstate; unlike the gluino, it couples the top of one chirality to the stop of the other. With the stops integrated out, the amplitudes are governed by the propagator matrix $D = (M_{\tilde{t}}^2)^{-1}$ in the $(\tilde{t}_R, \tilde{t}_L)$ basis. The absorptive part comes from the cut through $\tilde{g}\,\bar{t}$, and the CP-odd term is proportional to $m_{\tilde{h}} m_{\tilde{g}} f_2(m_{\tilde{g}}^2/m_{\tilde{h}}^2)\sin\phi_{12}$, where ϕ_{12} is the relative phase of the two Majorana masses. The two coupling structures contribute to this term with equal magnitude and opposite sign: a gaugino mass insertion enters with the phase $e^{i\phi}$ for a coupling to $\lambda\psi$ and $e^{-i\phi}$ for a coupling to $\lambda^\dagger\psi^\dagger$. For a pure higgsino they cancel. The asymmetry vanishes at this order for any stop masses, any stop mixing and any phase of X_t; we find $|\epsilon_{\mathrm{CP}}| < 10^{-18}$ over a scan of these parameters. A bino admixture a_B breaks the symmetry between the structures and gives

$$\epsilon_{\mathrm{CP}} \simeq 7\times10^{-4}\, a_B \qquad (m_{\tilde{t}} = 2\mu,\ X_t = 2.5\, m_{\tilde{t}},\ m_{\tilde{g}} = 0.4\mu,\ \sin\phi_{12} = 1,\ \lambda'' \gg g_3), \tag{21}$$

which reaches the required $(1.6\text{–}7)\times10^{-4}$ for $a_B \gtrsim 0.25$ (Fig. 3); for $m_{\tilde{t}} = 3\mu$ it requires $a_B \gtrsim 0.6$. The higgsino chain therefore generates the required asymmetry only for a well-tempered bino–higgsino state ($M_1 \simeq \mu$) with stops within a factor two or three of the higgsino mass, and $\mu > m_{\tilde{g}}$.

6.4 Other moduli stabilisation schemes

The entropy wall does not depend on the construction, only on m_Φ and the total width. Table 4 applies Eqs. (4) and (11) to the three classes of compactification studied in [22], using the decay coefficients $\Gamma = c\, m^3/M_{\mathrm{Pl}}^2$ listed there (IIB/KKLT: $c_{\mathrm{tot}} \simeq 0.13$, $m_\Phi = 10^6$–8×10^6 GeV; heterotic:

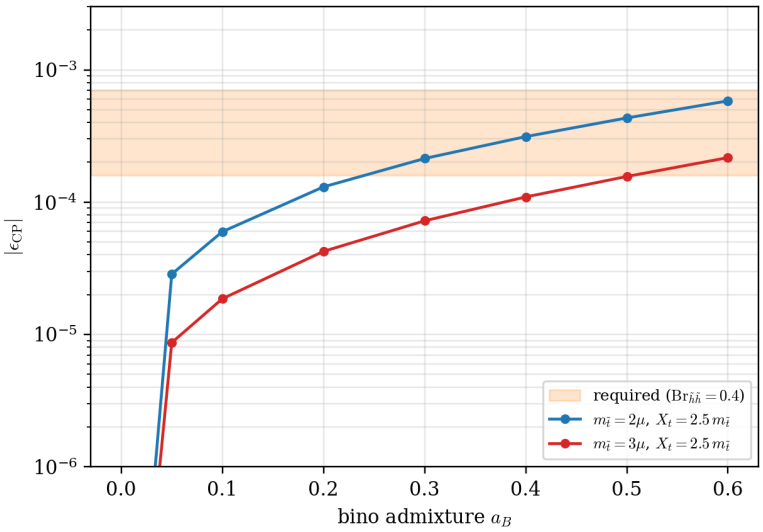


Figure 3: CP asymmetry in the decay of a neutralino, a mixture of the neutral higgsino and the bino with bino fraction a_B, to tsb through a gluino loop ($\sin\phi_{12} = 1$, $m_{\tilde{g}} = 0.4\mu$). The asymmetry vanishes for $a_B = 0$. The band is the asymmetry required in sequestered LVS for $\mathrm{Br}_{\tilde{h}\tilde{h}} = 0.4$.

$c_{\rm tot} \simeq 0.21$, $m_\Phi = 2$–6×10^6 GeV; G_2: $0.1 \le c_{\rm tot} \le 36$, $m_\Phi = 6 \times 10^4$–1.6×10^5 GeV), together with the sequestered LVS window of Sec. 6.3.

	T_R [GeV]	$\mathcal{D}(131.7\ \mathrm{GeV})$	η_L	$\epsilon_{\rm req}$	gaugino pairs
IIB / KKLT	0.16–3.0	10^{-16}–10^{-9}	10^{-14}–10^{-8}	$(3\text{–}7) \times 10^{-4}$	open
heterotic	0.5–2.5	10^{-13}–10^{-10}	10^{-11}–10^{-8}	$(2.7\text{–}4.5) \times 10^{-4}$	open
G_2	0.003–0.17	10^{-26}–10^{-16}	10^{-21}–10^{-13}	$(1\text{–}24) \times 10^{-4}$	open
sequestered LVS	0.4–54	10^{-14}–10^{-3}	10^{-10}–0.02	$(1.3\text{–}5.6) \times 10^{-4}$	closed

Table 4: Entropy-wall quantities in four moduli-dominated cosmologies. $\epsilon_{\rm req} = Y_B^{\rm obs}/Y_\Phi$ with $Y_\Phi = \frac{3}{4}T_R/m_\Phi$ is the baryon number per modulus decay required of baryon-violating decays. The last column states whether the modulus can decay to a gaugino pair: open in [22] for the first three, closed in LVS (Sec. 6.3).

In every case T_R lies below the 6.1 GeV required for electroweak-scale generation before the decay, which in these schemes translates into $m_\Phi \gtrsim 1.1$–1.4×10^7 GeV for $c_{\rm tot} \lesssim 0.2$ and 2×10^6 GeV for $c_{\rm tot} = 36$. A pre-decay asymmetry is therefore diluted by 10^{-9}–10^{-26}, and lepton-violating decays are converted with $\eta_L \lesssim 10^{-8}$ except in the upper part of the LVS window. What survives is baryon-violating decay, and the required baryon number per decay, $\epsilon_{\rm req} \sim 10^{-4}$–$10^{-3}$, is the same in all four constructions. It is a property of the entropy wall, not of the stabilisation scheme. LVS differs in whether the decay chain that would supply ϵ exists: the gaugino-pair channel used in [22] is open for the first three and closed for LVS.

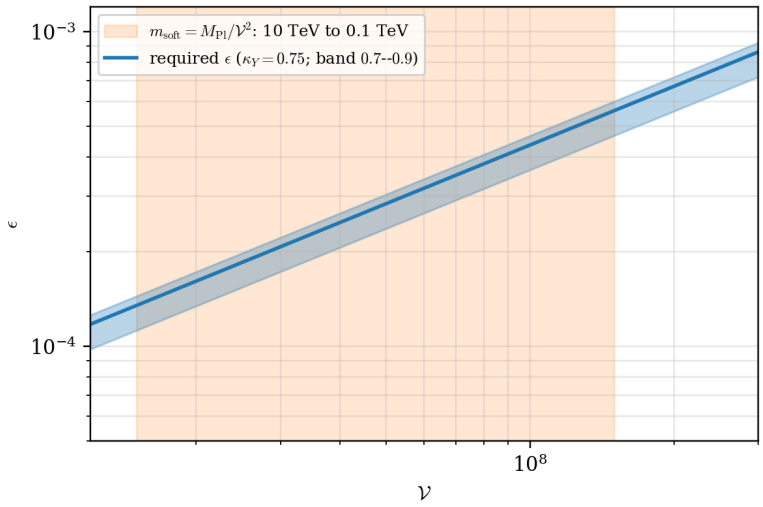


Figure 4: CP-violating baryon number per decay required to reproduce $Y_B^{\rm obs}$ from baryon-violating decays of the volume modulus in the explicit Swiss-cheese vacua, for $T_R \lesssim 100$ GeV where $\eta_B \simeq 1$.

Channel	Condition	$\mathcal{V}$ range
Pre-decay EW baryogenesis	$T_R \gtrsim$ 6–38 GeV	$\lesssim (1.7\text{–}3.8) \times 10^7$
Axion velocity, bulk/fibre	$F \lesssim 4 \times 10^{10}$ GeV	excluded ($F \simeq 9 \times 10^{15}$)
Axion velocity, blow-up	$\Gamma_\rho \gtrsim H_*$	excluded (fast decay)
Decay leptogenesis	$T_R \gtrsim$ 100–300 GeV	$\lesssim 1.1 \times 10^7$
B-violating decays	$\epsilon\, \eta_B \kappa_Y T_R/m \simeq Y_B^{\rm obs}$	$\lesssim 3.1 \times 10^{12} \epsilon^{4/3}$
B-violating decays, sequestered	$\epsilon \simeq (1\text{–}6) \times 10^{-4}$; gluino chain closed	1.5×10^7–1.5×10^8
Affleck–Dine, baryonic	see Sec. 7	$\lesssim 6{\times}10^5\, \zeta$; $\lesssim 10^4$ in the effective theory

Table 5: Summary of constraints on LVS, assuming a single modulus-dominated era and no thermal inflation.

7 Affleck–Dine baryogenesis

7.1 Setup

For a flat direction lifted by $W = \lambda\phi^n/(nM_*^{n-3})$,

$$V = (m_{\rm AD}^2 - cH^2)|\phi|^2 + \frac{\lambda^2|\phi|^{2n-2}}{M_*^{2n-6}} + \Big(\frac{(a\, m_{\rm AD} e^{i\alpha} + a_H H e^{i\alpha_H})\lambda\phi^n}{nM_*^{n-3}} + \text{h.c.}\Big). \tag{22}$$

For $H \gg m_{\rm AD}$ the field tracks $|\phi_0|^2 = \big(\sqrt{c/(n-1)}\, HM_*^{n-3}/\lambda\big)^{2/(n-2)}$ and begins to oscillate at $H = m_{\rm AD}$, leaving a charge density $n_B = \epsilon\, m_{\rm AD}|\phi_0|^2$. The relative phase $\delta = \alpha - \alpha_H$ is the CP-violating input. The udd direction has $n = 6$; LH_u has $n = 4$. The operator scale M_* is not fixed by the four-dimensional analysis; we write $M_* = \zeta M_{\rm Pl}$. Sec. 7.5 shows that the result depends sharply on ζ.

7.2 Yield

We consider $m_{\rm AD} > m_{\rm mod}$, the generic LVS ordering ($m_{\rm AD} \sim m_{3/2}$, $m_{\rm mod}/m_{3/2} \sim \mathcal{V}^{-1/2}$), with inflaton oscillations until $H \simeq m_{\rm AD}$ and reheating at $T_{\rm RH} \le T_{\rm AD}$, $3m_{\rm AD}^2 M_{\rm Pl}^2 = \frac{\pi^2}{30} g_* T_{\rm AD}^4$. The maximal yield is $Y_{\rm AD} = \epsilon|\phi_0|^2 T_{\rm AD}/4 m_{\rm AD} M_{\rm Pl}^2$; for lower $T_{\rm RH}$ it is reduced by $T_{\rm RH}/T_{\rm AD}$, so what follows are upper bounds. The modulus later oscillates at $H = m_{\rm mod}$ with energy fraction $r_i = \Delta^2/6M_{\rm Pl}^2$ and produces the dilution $\mathcal{D} = \frac{4}{3}\kappa_Y (T_R/T_i)(1-r_i)/r_i$, $T_i \propto \sqrt{m_{\rm mod} M_{\rm Pl}}$. For $n = 6$ the dependence on $m_{\rm AD}$ cancels, and

$$Y_B \simeq 0.13\, \eta\, \epsilon\, \zeta^{3/2} \sqrt{c_\Gamma}\, \frac{m_{\rm mod}}{M_{\rm Pl}} \qquad (r_i = \tfrac{1}{6},\ c = \lambda = 1,\ g_* = 106.75). \tag{23}$$

The coefficient is 0.24 for $g_R = 10.75$; the dependence on c and λ is weak ($c^{1/4}\lambda^{-1/2}$), while $Y_B \propto 1/r_i$, so $\Delta = M_{\rm Pl}$ is the conservative choice. Because $T_R \gg T_*$ in the relevant range, $\eta = 28/79$ for both baryonic and leptonic directions. For $\zeta = 1$, $\epsilon = 0.51$, $Y_B = Y_B^{\rm obs}$ requires $m_{\rm mod} \gtrsim 8.7 \times 10^9$ GeV.

7.3 Charge efficiency

We integrate the field equations of (22) in a matter-dominated background ($H = 2/3t$) in units $\phi = M_{\rm Pl} f$, $\tau = m_{\rm AD} t$, $h = H/m_{\rm AD}$. For $h \gg 1$ the equations have a scale-free attractor $f = R\, t^{-1/(n-2)} e^{i\theta_{\rm min}}$ with $n\theta_{\rm min} + \arg a_{\rm tot} = \pi$; starting on it with the exact velocity removes transients that otherwise change ϵ by a factor two. The comoving charge is normalised to $a^3 m_{\rm AD}|\phi_0|^2$ at $H = m_{\rm AD}$, which defines ϵ. For the benchmark $a = a_H = c = \lambda = 1$, $\delta = \pi/2$ we find $|\epsilon| = 0.51$ ($n = 6$) and 0.69 ($n = 4$). A scan of 162 points ($c, a, a_H \in \{0.3, 1, 3\}$, six values of δ) gives median $|\epsilon| = 0.16$, 90th percentile 0.55, maximum 1.2. The benchmark has maximal CP phase; the median is the more representative value.

7.4 Results for $M_* = M_{\rm Pl}$

For $m_{\rm AD} = 0.1\, m_{3/2}$, $r_i = \frac{1}{6}$, $c_\Gamma = 1$, the maximal volume for which $Y_B^{\rm obs}$ is reached is, for $\epsilon = 1.2$, 0.51, 0.16:

	$\epsilon = 1.2$	0.51	0.16
udd, Swiss-cheese	1.1×10^6	5.9×10^5	2.8×10^5
udd, K3-fibred	2.5×10^5	1.5×10^5	7.3×10^4
LH_u, Swiss-cheese	1.9×10^4	1.2×10^4	6.9×10^3

At $\mathcal{V} = 5.9 \times 10^5$, $m_{3/2} = 4.1 \times 10^{12}$ GeV, $m_{\rm mod} = 8.7 \times 10^9$ GeV and $T_R = 7.9 \times 10^4$ GeV. The leptonic direction fails because $|\phi_0|^2$ is smaller than for $n = 6$ by $0.87\sqrt{m_{\rm AD}/M_{\rm Pl}} \sim 10^{-4}$–$10^{-3}$. If the soft terms are sequestered below the modulus mass ($m_{\rm AD} < m_{\rm mod}$), the yield is $\propto T_R/\sqrt{m_{\rm AD} M_{\rm Pl}}$ and the window extends to $\mathcal{V} \sim 10^7$ for $m_{\rm AD}/m_{\rm mod} \sim 10^{-6}$.

7.5 Consistency of the effective theory

In LVS the string and Kaluza–Klein scales are $M_s \simeq g_s^{1/4} M_{\rm Pl}/\sqrt{4\pi\mathcal{V}}$ and $M_{\rm KK} \sim M_{\rm Pl}\mathcal{V}^{-2/3}$; at $\mathcal{V} = 5.9 \times 10^5$, $M_s \approx 7 \times 10^{14}$ GeV and $M_{\rm KK} \approx 3 \times 10^{14}$ GeV (up to $O(1)$ convention-dependent factors). With $m_{\rm AD} = 0.1\, m_{3/2}$,

$$\frac{T_{\rm AD}}{M_s} \simeq 0.8, \qquad \frac{|\phi_0|}{M_s} \simeq 2.2\, \mathcal{V}^{1/4}\, \zeta^{3/4} \qquad (n = 6). \tag{24}$$

The reheating temperature at which Eq. (23) is saturated lies at the string scale for every $\mathcal{V}$, and for $\zeta = 1$ the flat direction is displaced to $\approx 60\, M_s$ at $\mathcal{V} = 5.9 \times 10^5$. The four-dimensional analysis is then outside its range of validity. The charge efficiency ϵ is independent of ζ, since ζ can be absorbed into λ in Eq. (22); ζ enters only through $|\phi_0|^2 \propto \zeta^{3/2}$. Evaluating the full yield, Eq. (23) without the approximations of the coefficient, in the explicit vacua (and in the K3 vacua with the fibre modulus) gives Table 6. The scan reproduces the values for $\zeta = 1$ above to within 6%, and $\mathcal{V}_{\max} \propto \zeta$ to within 15%. Three prescriptions for M_* are of interest: $\zeta = 1$; M_* chosen so that $|\phi_0| = M_s$; and $M_* = M_s$.

	Swiss-cheese			K3-fibred		
ϵ	1.2	0.51	0.16	1.2	0.51	0.16
$\zeta = 1$	1.0×10^6	6.0×10^5	2.8×10^5	2.4×10^5	1.5×10^5	7.3×10^4
$\zeta = 0.1$	1.1×10^5	6.4×10^4	3.0×10^4	3.1×10^4	1.8×10^4	9.2×10^3
$\lvert\phi_0\rvert = M_s$	1.6×10^4	1.1×10^4	6.0×10^3	6.7×10^3	4.5×10^3	2.7×10^3
$M_* = M_s$	4.0×10^3	2.8×10^3	–	2.0×10^3	1.4×10^3	–

Table 6: Maximal volume for baryonic Affleck–Dine baryogenesis ($n = 6$, $m_{\rm AD} = 0.1\, m_{3/2}$, $c_\Gamma = 1$, $r_i = \frac{1}{6}$) as a function of the operator scale and the charge efficiency. A dash indicates no solution for $\mathcal{V} \geq 2.5 \times 10^3$ (Swiss-cheese) or 10^3 (K3).

Within the effective theory the maximal volume is $\mathcal{V} \sim 10^4$ (Swiss-cheese) and 5×10^3 (K3-fibred), and for $M_* = M_s$ it is $\sim 3 \times 10^3$ and 1.4×10^3, where the large-volume expansion is no longer under control.

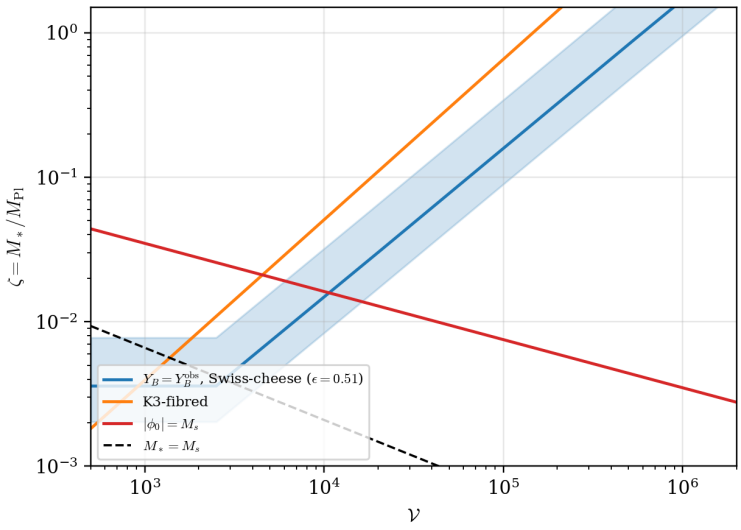


Figure 5: Baryonic Affleck–Dine baryogenesis in the plane of volume and operator scale $\zeta = M_*/M_{\rm Pl}$. The observed asymmetry is reached above the blue line (Swiss-cheese, $\epsilon = 0.51$; band: $\epsilon = 0.16$–1.2) or the orange line (K3-fibred); the effective theory is valid below the red line, $|\phi_0| = M_s$. The dashed line is $M_* = M_s$.

The allowed region is shown in Fig. 5. Several further effects bear on the analysis: (i) the blow-up modulus and its axion have $m \simeq (17$–$78) m_{3/2} \gg m_{\rm AD}$, so they begin to oscillate before $H = m_{\rm AD}$ and decay at $H \sim 10^4$ GeV, long after, and may dominate the energy density across the epoch in which the flat direction is displaced; Case I assumes inflaton-driven expansion. (ii)

The condensate can fragment into Q-balls; this changes the dynamics but not the total charge. (iii) With $m_{\rm AD} \gtrsim 10^9$ GeV the low-energy theory is not supersymmetric and dark matter must be supplied separately. (iv) The sequestered case is more favourable, since $|\phi_0| \propto m_{\rm AD}^{1/4}$, but the analysis of Hubble-induced terms during modulus domination is beyond the scope of this paper.

The conclusion is that baryonic Affleck–Dine baryogenesis with gravity-mediated soft terms does not give a controlled window in LVS at large volume; it is possible at $\mathcal{V} \lesssim 10^4$ (K3: 5×10^3) or with sequestered soft terms. Affleck–Dine baryogenesis in LVS was studied in [20]; the present analysis concerns the case in which the modulus itself dilutes the asymmetry, as in [19].

8 Discussion

Within the assumptions stated (a single modulus-dominated era, no thermal inflation), electroweak-scale baryogenesis before the decay, axion-velocity mechanisms with bulk, fibre or blow-up axions, and leptogenesis from modulus decay are excluded in LVS at the volumes that give TeV-scale supersymmetry. What remains is baryon-violating decay, which for sequestered soft terms and TeV-scale superpartners at $\mathcal{V} \sim 10^7$–10^8 requires $\epsilon \sim 10^{-4}$–10^{-3}, through the Higgs and higgsino channels since the gaugino channel is closed, where the gluino-loop asymmetry vanishes for a pure higgsino and reaches the required size only for a bino admixture of at least 25%, and Affleck–Dine baryogenesis, whose controlled regime is small volume or sequestered soft terms. The sphaleron results of Sec. 3 apply to any modulus-dominated cosmology, and the required baryon number per decay, $\epsilon \sim 10^{-4}$–10^{-3}, is common to the IIB, heterotic, G_2 and LVS constructions (Table 4).

Limitations. (1) The lattice parametrisation is discontinuous at T_c and c is calibrated; efficiencies are uncertain at $O(1)$ through c, though the exponent in $\eta_L \propto T_R^4$ is robust. (2) $g_*(T)$ is approximated near the QCD crossover. (3) The Swiss-cheese vacua are AdS and use $W_0 = A = 1$; c_Γ is a parameter, and varying it over $[0.3, 8]$ changes $\mathcal{V}_{\rm max}$ by a factor $\lesssim 2$. (4) The K3 loop potential is conjectural [9]; variations in the loop data change $\mathcal{V}_{\rm max}$ by a factor $\lesssim 4$. (5) All bounds assume the lightest modulus dominates (Sec. 2) and, for $T_R \lesssim 1$ GeV, a specific pre-oscillation history. (6) The kinetic bound assumes Chern–Simons-driven baryogenesis; non-sphaleron axion mechanisms are not constrained. (7) ϵ for baryon-violating decays and the string-theoretic origin of the A-terms are not derived.

Data and code. All numerical results are generated by the code at `https://github.com/vishal1601-2005/Entropy-walls-LVS` (MIT licence), which contains the intermediate data and the LaTeX source. The inputs are the lattice rate [18], the K3 parameter sets [9] and $Y_B^{\rm obs}$.

A Dilution coefficient

Let the sudden-decay entropy be $S_{\rm fin} = \frac{2\pi^2}{45} g_R T_R^3 a_d^3$, with $\rho_\phi(a_d) = 3M_{\rm Pl}^2\Gamma^2$. Since $\rho_\phi \propto H^2$ and $\rho_\phi \propto a^{-3}$, $a_d^3/a_{\rm gen}^3 = H_{\rm gen}^2/\Gamma^2$, with $H_{\rm gen}$ from Eq. (3). Hence

$$\mathcal{D} = \frac{g_*}{g_R}\Big(\frac{T_{\rm gen}}{T_R}\Big)^3\Big(\frac{\Gamma}{H_{\rm gen}}\Big)^2, \qquad \frac{\Gamma}{H_{\rm gen}} = \frac{b}{\kappa}\Big(\frac{T_R}{T_{\rm gen}}\Big)^4, \qquad \frac{b}{\kappa} = \frac{36\,b^2}{\pi^2 g_*} = 0.4\,\frac{g_R}{g_*}, \tag{25}$$

with $b = \sqrt{\pi^2 g_R/90}$. Therefore $\mathcal{D} = 0.16\,(g_R/g_*)(T_R/T_{\rm gen})^5$, which is Eq. (4). The same ratio gives $\Gamma/H(T) = \frac{2g_R}{5g_*}(T_R/T)^4$ used in Sec. 3.3.

B Susceptibility

In units of $T^2/6$ per species, with flavour-universal potentials, hypercharge neutrality reads $n_g[\mu_Q + 2\mu_u - \mu_d - \mu_L - \mu_e] + 2\mu_H = 0$ (the Higgs term carries the bosonic factor). With $\mu_u = \mu_Q + \mu_H$, etc., this gives $\mu_H = \frac{3}{7}(\mu_L - \mu_Q)$. Then $B = 12\mu_Q$, $B - L = 12\mu_Q - 9\mu_L + 3\mu_H$, and $\mu_{\rm CS} = 9\mu_Q + 3\mu_L$. For $B - L = 0$, $\mu_L = \frac{25}{18}\mu_Q$ and $\mu_Q = \frac{6}{79}\mu_{\rm CS}$, so $n_B = \frac{12}{79}T^2\mu_{\rm CS}$. For $\mu_{\rm CS} = 0$, $\mu_L = -3\mu_Q$ and $B/(B-L) = 28/79$.

C Kähler metric of the blow-up modulus

For $K = -2\ln\mathcal{V}$ with $\mathcal{V} = \gamma(\tau_b^{3/2} - \tau_s^{3/2})$ and $\tau = \frac{1}{2}(T + \bar{T})$, $K_{s\bar{s}} = \frac{1}{4}\partial^2_{\tau_s}K$. For $\tau_s^{3/2} \ll \tau_b^{3/2}$, $K = -2\ln(\gamma\tau_b^{3/2}) + 2\tau_s^{3/2}/\tau_b^{3/2} + \ldots$, hence $K_{s\bar{s}} \simeq 3\gamma/8\sqrt{\tau_s}\mathcal{V}$. We checked this against the exact second derivative at $\mathcal{V} = 1.5\times10^{14}$, $\tau_s = 25.1$ (four digits).

D Explicit amplitude check of the CP asymmetry

With the stops integrated out, the interactions are

$$\mathcal{L}_{\rm eff} = \sum_{\alpha\beta} D_{\alpha\beta}\, J^*_\alpha J_\beta, \qquad J_\alpha = \sum_{i=1,2}\Big[A_{i\alpha}(\lambda_i\psi_Q) + B_{i\alpha}(\lambda_i^\dagger\psi^\dagger_{U^c})\Big] + \delta_{\alpha R}\,\rho\,(\psi_{b^c}\psi_{s^c}), \tag{26}$$

with $\alpha \in \{R, L\}$, $D = (M_{\tilde{t}}^2)^{-1}$, and $\lambda_{1,2}$ the Majorana fermions. For the gluino $A_L = -\sqrt{2}g_3$, $B_R = \sqrt{2}g_3$; for the neutral higgsino $A_R = B_L = -y_t/\sqrt{2}$; for a bino admixture $B_R = \frac{2\sqrt{2}}{3}g_1 a_B$ and $A_L = -\frac{\sqrt{2}}{6}g_1 a_B$. The tree amplitudes for $\lambda_1 \to tsb$ and $\bar{t}\bar{s}\bar{b}$, the three-body amplitude for $\lambda_1 \to t\lambda_2\bar{t}$ (two diagrams, since λ_2 is Majorana) and the amplitude for $\lambda_2\bar{t} \to bs$ are evaluated with explicit two-component spinors. The absorptive part follows from unitarity,

$$\mathcal{T}^{\rm abs}_{fi} = \tfrac{i}{2}\sum_n \int d\Phi_n\, \mathcal{T}_{fn}\mathcal{T}_{ni}, \qquad n = (\lambda_2, t, \bar{t}), \tag{27}$$

with the top a spectator, and the asymmetry is $\Gamma(f) - \Gamma(\bar{f})$ integrated over the three-body phase space. The relative sign of the f and $\bar{f}$ cuts is a fermion-ordering sign; with the opposite sign the asymmetry vanishes identically, and the sign used reproduces [22] for right-handed couplings.

The implementation reproduces the tree rate (39) of [22] for all stop mixings and couplings tested, and the cross-chirality terms of (40). For right-handed couplings it reproduces the f_2 term of (41), with the dependence on $x = m^2_{\tilde{\lambda}_2}/m^2_{\tilde{\lambda}_1}$ confirmed at $x = 0.09$, 0.25, 0.49 and the dependence $\sin\phi_{12}$ on the phase. It differs from (41) in two respects. First, the left-handed term $\kappa^2_{1L}\kappa^2_{2L}c_t^2 s_t^4$ enters with the opposite sign to the right-handed one. Second, there is no f_1 term. Both follow from rephasing invariance: a term without a gaugino mass insertion cannot depend on the phase of the gaugino mass, and the right- and left-handed mass-insertion terms carry opposite phases. The same-chirality three-body rate differs from (40) by 3% at $x = 0.25$. Neither difference affects the right-handed benchmark (47).

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